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MILITARY SPECIFICATION

ADAPTER, PRESSURE FUEL SERVICING, AIRCRAFT, GENERAL SPECIFICATION FOR

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers adapters used as airplane fuel connections for use in conjunction with pressure fuel servicing equipment, for hydrocarbon fuels only.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

P-S-661 — Solvent, Dry Cleaning

QQ-C-320 — Chromium, Plating, (Electrodeposited)

QQ-P-416 — Plating, Cadmium, (Electrodeposited)

UU-P-271 — Paper, Wrapping, Waterproofed, Kraft.

PPP-B-566 — Boxes, Folding, Paperboard

PPP-B-585 — Boxes, Wood, Wirebound

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated

PPP-B-601 — Boxes, Wood, Cleated-Plywood

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner

PPP-B-636 — Boxes, Fiber

PPP-B-676 — Boxes, Set-Up, Paperboard

PPP-T-60 — Tape; Pressure Sensitive Adhesive, Waterproof-for Packaging and Sealing.

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MIL-B-138 — Boxes, Wood, Fiberboard-Lined For Overseas Shipment (For Weight Contents Not Exceeding 500 Pounds)

MIL-S-3136 — Standard Test Fluids Hydrocarbon and Iso-Octane

FSC 1560

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MIL-B-4229 — Boxes, Paperboard, Metal-Stayed

MIL-D-5028 — Drawings and Data Lists: Preparation of Manufacturers' (for Production Aeronautical and Associated Equipment)

MIL-J-5161 — Jet Fuel, Referee

MIL-P-5315 — Packing, "O" Ring, Hydrocarbon Fuel Resistant

MIL-G-5572 — Gasoline, Aviation: Grade 80/87, 91/96, 100/130, 115/145

MIL-J-5624 — Jet Fuel, Grades JP-3, JP-4, and JP-5

MIL-N-5877 — Nozzle, Pressure Fuel Servicing, Locking, Type D-1

MIL-E-7729 — Enamel, Gloss, Aircraft Application

MIL-S-7742 — Screw Threads, Standard, Aeronautical

MIL-C-8068 — Cloth, Cloated, Nylon, Rubber-Coated, Fuel Resistant

MIL-P-8585 — Primer; Zinc Chromate, for Aircraft use

MIL-A-8625 — Anodic Coatings, for Aluminum and Aluminum Alloys

MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper Overlaid

MIL-L-10547 — Liners, Case, Waterproof

MIL-N-25027 — Nut, Self-Locking, 250°F, 550°F, and 800°F

STANDARDS

MILITARY

MIL-STD-129 — Marking for Shipment and Storage

MIL-STD-130 — Identification Marking of U.S. Military Property

MS29512 — Gasket—Fuel Resistant Straight Thread Tube Fitting, Boss

MS29513 — Packing "O" Ring Hydrocarbon Fuel Resistant

MS29520 — Nozzle, Pressure Fuel Servicing Locking, Type D-1

MS33540 — Safety Wiring—General Practices For

MS33586 — Metals, Definition of Dissimilar

MS33588 — Nuts and Plate Nuts, Self-Locking, Functional Limitations of

MS20995 — Wire, Lock

AIR FORCE-NAVY AERONAUTICAL

AND10049 — Bosses — Fluid Connection, Internal Straight Thread

PUBLICATIONS

AIR FORCE-NAVY AERONAUTICAL BULLETINS

No. 143 — Specifications and Standards; Use of[1]

No. 438 — Age Controls for Synthetic Rubber Parts

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

[1] Applicable only to Air Force and Navy purchases.

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3. REQUIREMENTS

3.1 Qualification. The adapters furnished under this specification shall be a product which has been tested and has passed the Qualification tests specified herein.

3.2 Materials. Materials and processes used by the manufacturer of adapters shall be suitable for the purpose, and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used provided the specifications are released by the Services and contain provision for adequate tests. The use of contractor's specifications will not constitute waiver of Government inspection. All materials used in the adapter shall be sufficiently resistant to fuel conforming to Specification MIL-J-5161, MIL-S-3136, MIL-G-5572, or MIL-J-5624, of aromatic content from 0 to 30 percent, to assure satisfactory operation as herein defined.

3.2.1 *Metals.* All metals used in the construction of adapters shall be of a corrosion-resistant type or shall be suitably protected to resist corrosion. The use of dissimilar metals shall be avoided whenever practicable or shall be in accordance with Standard MS33586. The use of magnesium or any alloy thereof is prohibited.

3.2.2 *Castings.* Castings shall be of high quality, clean, sound, and free from blowholes, porosity, cracks, and any other defects.

3.2.3 *Selection of materials.* Except for Army procurement, specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected for Air Force and Navy purchases in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.2.3.1 *Standard parts.* AN or MS standard parts shall be used whenever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts, such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the AN or MS standard parts without alteration, and provided the corresponding AN or MS part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding AN or MS standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design. The adapters shall be so designed that a means shall be provided for connecting an aircraft pressure fuel servicing nozzle conforming to Standard MS29520 to the airplane fuel system for the purpose of fueling or defueling the system. The assembly shall so operate that when the nozzle is secured to the adapter and is actuated from the "nonflow" to the "flow" position, the adapter sealing unit will operate from the "closed" to the "open" position. Emphasis shall be placed on eliminating any tendency for the sealing unit to cock or "hang-up" in such a manner that it does not properly close and seal when the nozzle is actuated from the "flow" to the "nonflow" position.

3.3.1 *Sealing provision.* The adapter shall incorporate a means of sealing when the nozzle is not connected. The sealing unit shall be operable by the nozzle action.

3.3.2 *Lubricants.* The adapter shall require no lubricant throughout its life.

3.3.3 *"O"-Ring packings.* "O"-ring packings, wherever used, shall conform dimensionally to Standard MS29512 or MS29513, as applicable, and physically to Specification MIL-P-5315.

3.3.4 *Synthetic rubber parts.*

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3.3.4.1 *Marking.* All synthetic rubber parts, except "O" rings and parts with no suitable surface, shall have painted, stamped with ink, or otherwise noted on the part, the year and quarter of the curing date of the part.

3.3.4.2 *Age controls for synthetic rubber parts.* Age controls for adapters shall be in accordance with the requirements of ANA Bulletin No. 438. The markings shall be located on an exposed surface not normally in contact with fuel.

3.3.4.3 *Serviceability.* All synthetic rubber parts shall be readily replaceable with a minimum replacement of attaching parts.

3.3.4.4 *Uniformity.* For adapters which include parts fabricated of synthetic material in contact with fuel, manufacturers shall control subsequent batches of the material, to provide for uniformity.

3.3.5 *Maintenance.* The adapter shall require no special tools or fixtures for overhaul or repair of the assembly. The adapter shall be so designed that all parts that may require replacement can be easily replaced without removing the adapter from the airplane.

3.3.6 *Total pressure loss.* The allowable loss of the total pressure shall be as specified in the applicable detail specification.

3.3.7 *Operating force.* The adapter sealing unit shall be operable by a nozzle conforming to Standard MS 29520, with forces not exceeding 50 pounds with the adapter dry or wet with fuel and with all flows and pressures specified in the applicable detail specification.

3.3.8 *Strength.* The adapter shall be able to withstand the following loads:

(a) An axial bending couple of 1,500 inch-pounds with the force applied in any direction normal to the adapter centerline.

(b) A combined load of 165 pounds axially and 600 inch-pounds couple applied in any plane containing the adapter axis.

3.3.9 *Sump drain provisions.* The sump drain boss shall conform to Drawing AND-10049 and shall be designed to leave a minimum of trapped fluid when the adapter axis is vertical.

3.3.10 *Electrical conductivity.* Complete electrical contact between the nozzle and adapter assemblies shall be established through the attaching lugs when the units are connected; and between the attaching lugs of the adapter and the adapter body mounting flange. The resistance between these points shall not exceed 10 ohms.

3.4 **Construction.** The adapter shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service use.

3.5 **Interchangeability.** All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL–D–5028.

3.6 **Dimensions.** The over-all dimensions and tolerances shall be as specified in the detail specification.

3.7 **Weight.** The weight of the adapter shall not exceed that specified on the applicable MS standard.

3.8 **Threads.** All screw threads shall conform to Specification MIL–S–7742.

3.9 **Locking of parts.** All threaded parts shall be positively locked by safety wiring, by self-locking nuts conforming to Specification MIL–N–25027, cotter pins, or other ap-

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proved methods. Safety wire when used shall be installed in accordance with Standard MS33540 and shall conform to Standard MS20995. (Self-locking nuts shall be used in accordance with Standard MS33588.) Self-locking nuts shall not be used where loosening or disengagement of the nut could result in the nut or other parts entering the fuel system. The use of lockwashers or staking is not permitted.

3.10 **Performance.** The adapters shall satisfy the performance requirements specified in 4.5 when subjected to the applicable tests specified therein.

3.11 **Finish.**

3.11.1 *Anodizing.* All aluminum-alloy parts shall be anodized in accordance with Specification MIL–A–8625, or adequately treated in some other acceptable manner for corrosion prevention.

3.11.2 *Protective treatment.* Steel parts, other than corrosion-resisting steel, not in moving contact, shall be cadmium plated in accordance with Specification QQ–P–416, type II. Sliding or rotating parts shall be chromium plated in accordance with Specification QQ–C–320 or adequately treated in some other acceptable manner for corrosion prevention, where galling of cadmium plating would cause malfunctioning. Brass, bronze, and copper-alloy parts shall be cadmium plated, or adequately treated in some other acceptable manner for corrosion prevention. Stainless steel parts shall be passivated.

3.11.3 *Paint finish.* Any parts painted shall be finished with one coat of primer conforming to Specification MIL–P–8585, and one coat of red enamel conforming to Specification MIL–E–7729.

3.12 **Identification of product.** Equipment assemblies, and parts shall be marked for identification in accordance with Standard MIL–STD–130.

3.13 **Workmanship.** Adapters shall be free of material defects such as folds, laps, seams, and other defects. Burrs and sharp edges are not acceptable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 *Classification of tests.* The inspection and testing of adapters shall be classified as follows:

(a) Qualification tests (4.2)

(b) Acceptance tests (4.3)

4.2 **Qualification tests.**

4.2.1 *Qualification test samples.* The Qualification test samples shall consist of 4 adapters upon which qualification is desired. Two of the adapters shall have been subjected to the Qualification tests by the contractor. The remaining 2 adapters shall be untested units. Samples shall be accompanied by 2 complete sets of detail and assembly drawings and a complete test report showing results of manufacturer's tests. Samples shall be identified as required and forwarded to the activity designated in the letter of authorization from the activity responsible for qualification.

4.2.1.1 *Manufacturer's drawings.* Manufacturer's drawings submitted with the Qualification test samples shall show a cutaway section showing all parts in their normal as-

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sembled position and shall specify part numbers of all parts and subassemblies. The following data shall be furnished on or together with the assembly drawing:

(a) Mounting dimensions

(b) Over-all dimensions

(c) Location, size, and type of plumbing connections

(d) Materials and construction, treatment and finish

(e) Pressure and flow rating

4.2.1.2 *Manufacturer's test report.* The test reports submitted with the Qualification test samples shall include the following:

(a) Report of all tests, graphically presented when practicable together with a detailed statement indicating conformance or extent of conformance with all requirements of this specification and the detail specifications, referring specifically to paragraph numbers. Wherever a requirement is considered to be not applicable, the report should so state.

(b) Summary of Endurance test 4.5.6

(c) Diagrams of all test setups

(d) Outline and description of test and test conditions

(e) Copies of test log sheets

(f) Photographs when available.

4.2.2 *Tests.* The Qualification test shall consist of the following tests, as described under "Test methods," and shall be conducted in the order listed:

Sample No. 1

(a) Inspection	4.5.1
(b) Calibration 4.5.2 (except 4.5.2.1.1 and 4.5.2.3.1)	
(c) Fuel resistance and extreme temperature	4.5.3
(d) Strength	4.5.5
(e) Tank leakage	4.5.4
(f) Disassembly and inspection	4.5.12

Sample No. 2

(a) Inspection	4.5.1
(b) Calibration 4.5.2 (except 4.5.2.1.1 and 4.5.2.3.1)	
(c) Endurance	4.5.6
(d) Accelerated corrosion	4.5.7
(e) Vibration	4.5.8
(f) Proof pressure	4.5.9
(g) Burst pressure	4.5.10
(h) Contaminated-fuel endurance	4.5.11
(i) Disassembly and inspection	4.5.12

4.3 **Acceptance tests.** The acceptance tests shall consist of Individual tests and a Sampling test.

4.3.1 *Individual tests.* Each adapter submitted for acceptance under contract shall be subjected to the following tests, as described under "Test methods," in the order specified:

(a) Inspection	4.5.1
(b) Functional (Acceptance test only)	4.5.2.1.1
(c) Leakage (Acceptance test only)	4.5.2.3.1
(d) Tank leakage	4.5.4

4.3.2 *Sampling tests.* One adapter shall be selected from each lot and subjected to the following test:

(a) Burst pressure test	4.5.10

4.3.2.1 *Lot.* A lot shall consist of not more than 200 adapters of the same part number,

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manufactured under the same conditions and submitted for acceptance at the same time.

4.3.3 *Rejection and retest.* Items which have been rejected may be reworked or replaced to correct the defects and resubmitted for acceptance. When a representative sample fails to meet the Sampling test, the lot represented shall be rejected. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.4 **Test conditions.** Unless otherwise specified by the procuring activity the following test conditions shall apply during the tests performed in accordance with this specification.

4.4.1 *Cleaning.* Prior to testing the adapter, all internal parts normally in contact with the fuel shall be thoroughly cleaned to remove all foreign matter.

4.4.2 *Test fluid.* Unless otherwise specified, fuel, in accordance with Specification MIL–G–5572, grade 100/130 or grade 115/145, or Specification MIL–S–3136, type I shall be used for all tests. Any fluid conforming to Specification P–S–661 (or any fluid acceptable to the procuring activity) may be used as a substitute for the test fuel for all tests except wherein a specific fluid is specified.

4.4.3 *Ambient temperature and pressure.* Unless otherwise specified, all tests shall be conducted with the adapter and fuel at a room temperature between 60° and 90°F and at atmospheric pressure.

4.4.4 *Test nozzle.* The test nozzle used shall conform to the requirements of Specification MIL–N–5877 or shall be a simulated nozzle conforming to figure 1.

4.4.5 *Test connections.* Unless otherwise specified, all tests shall be conducted without any part (such as the cap or nozzle) being connected to the adapter locking flange.

4.5 **Test methods.**

4.5.1 *Inspection.* The adapter shall be carefully and thoroughly examined to determine conformance to applicable drawings and the requirements of this specification which are not covered by specific test methods.

4.5.2 *Calibration.*

4.5.2.1 *Functional (Qualification test).*

(a) The test nozzle shall be inserted and locked into the adapter and the adapter sealing valve actuated a minimum of five times without any fuel flow.

(b) Test fluid shall be passed through the nozzle and adapter with a flow of 100 gpm with a minimum entrance static pressure of 40 psi under flow conditions and 50 psi in the shutoff position. The adapter sealing valve shall be actuated to open and shut a minimum of three times.

(c) Test outlined in (b) above shall be repeated at 200, 300, 400, 500, and 600 gpm flows.

(d) There shall be no evidence of chatter or binding and excessive force shall not be required to actuate the adapter sealing valve at any time during these tests.

4.5.2.1.1 *Functional (Acceptance test).* Each adapter shall be subjected to the test specified in 4.5.2.1 except that when test fluid is passed through the adapter, the flow need not be of any specified values as long as the specified pressures are used.

4.5.2.2 *Total pressure loss.* Using a test setup as specified in the detail specification and using the test nozzle, the test shall be

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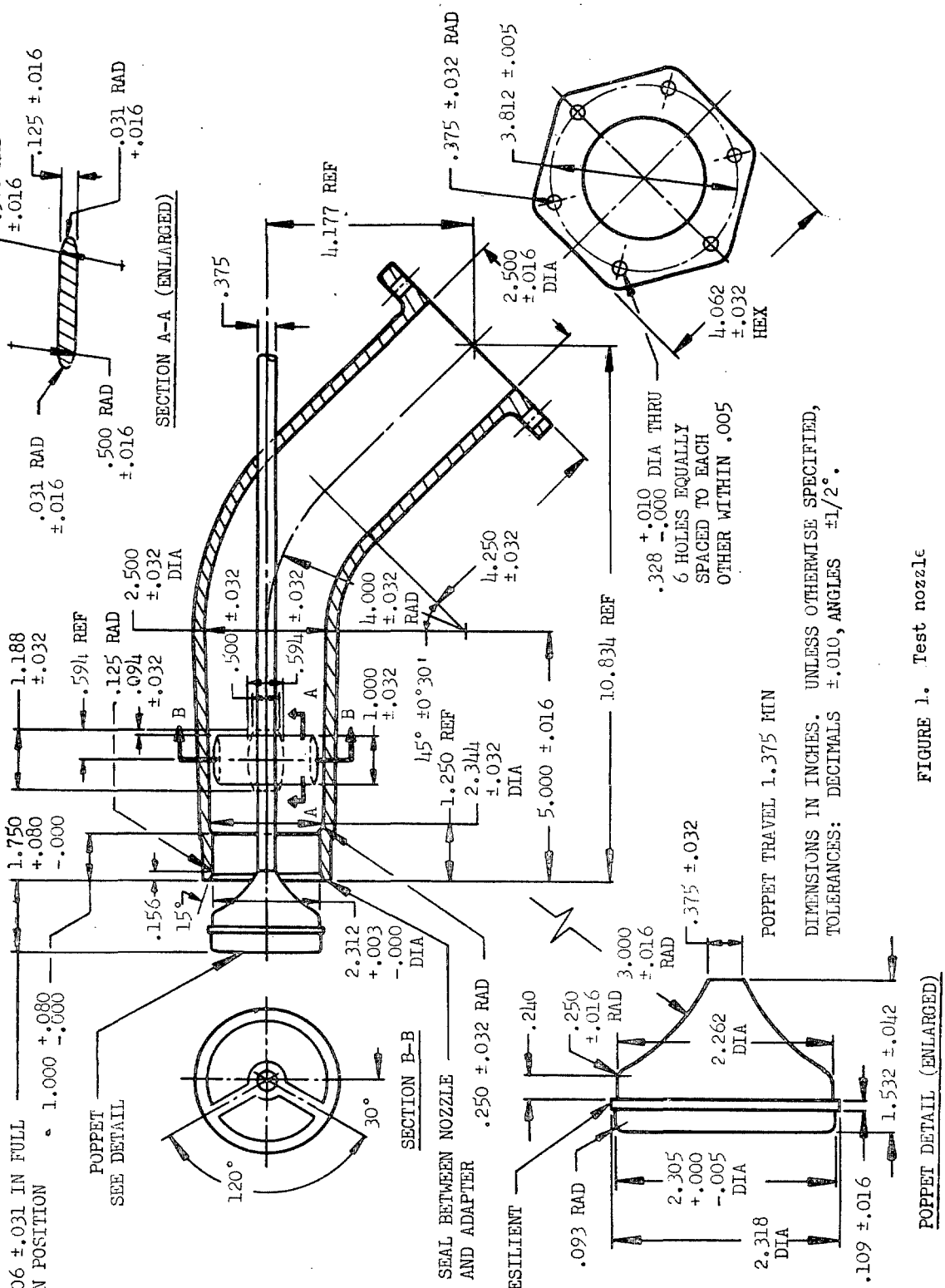


FIGURE 1. Test nozzle

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conducted with flows from 0 to 600 gpm. Sufficient data shall be taken to satisfactorily plot a "total pressure loss vs flow" curve. This curve shall accompany the Qualification test samples when submitted to the procuring activity. The curve shall show results for both the test fluid used and the corrected results for the reference fuel specified in the detail specification.

4.5.2.3 *Leakage (Qualification test).* The Leakage test shall be conducted over a fuel pressure range in ½-psi increments from —1 to +5 psi and in 10-psi increments from 10 to 60 psi. The pressure shall be applied to the adapter outlet. There shall be no evidence of leakage.

4.5.2.3.1 *Leakage (Acceptance test only).* Each adapter shall be subject to the test specified in 4.5.2.3 except that test pressures of —1, +5, +60, and +120 psi only shall be used.

4.5.3 *Fuel resistance and extreme temperature.* The Fuel resistance and Extreme temperature tests shall be conducted in accordance with table I.

4.5.4 *Tank leakage.* In a setup similar to figure 2 or 3, as applicable, the adapter shall be subjected to test fluid pressures of 5 and 25 psi for 1 minute each. There shall be no evidence of leakage at either pressure.

4.5.5 *Strength.* With the adapter retained by its mounting flange only, the test nozzle connected to the adapter, and with a fluid pressure of 60 psi imposed on the nozzle-adapter combination, a load as shown in figure 4 shall be applied intermittently for 1,000 cycles. There shall be no evidence of leakage, deterioration, fracture, or malfunction of any part of the adapter.

4.5.6 *Endurance.* Using a nozzle and test setup similar to the Functional test (4.5.2.1), the following procedure shall be followed:

(a) The nozzle shall be inserted and locked into the adapter.

(b) The nozzle shall be actuated from the "nonflow" to the "flow" position a total of four times with a fluid pressure of approximately 50 psi applied to the nozzle when in the "nonflow" position.

(c) The nozzle shall be removed from the adapter.

(d) Repeat (a), (b), and (c) a minimum of 2,500 nozzle engagements, which will subject the adapter sealing valve to a minimum of 10,000 cycles of operation. There shall be no evidence of deterioration to cause malfunction of the adapter. The leakage test (4.5.2.3) shall be reconducted at this point.

4.5.7 *Accelerated corrosion.* The adapter shall be immersed in a solution consisting of 2½ percent by weight of NaCl in distilled water. After immersion, the solution shall be drained and the adapter shall be heated in an oven to a temperature of 130° ± 5°F for a period of not less than 1 hour. The immersion and heating cycle shall be repeated 50 times. The adapter shall not be operated at any time during the immersion and heating cycle. Immediately after completion of the immersion and heating cycles, the adapter shall be washed out with warm water to remove all salt accumulations, after which, the valve shall be dried, wetted with fuel, and operated for 3 complete cycles (see 6.2.1). Corrosion of any part of the adapter to a degree which might affect performance shall be cause for rejection.

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TABLE I. *Fuel resistance and Extreme temperature test schedule*

Test period	Fuel resistance					Extreme temperatures	
	Initial drying	Phase I Soak	Phase I Dry	Phase II Soak	Phase II Dry	Low temperature	High temperature
Component configuration	Drained, valve closed, outlet port open	Valve closed, outlet port open	Drained, valve closed, outlet port open	Valve closed, outlet port open	Drained, valve closed, outlet port open	Valve closed, outlet port open	Valve closed, outlet port open
Test fluid	None	MIL-S-3136, type III	None	MIL-S-3136 type I	None	MIL-S-3136, type I	MIL-S-3136, type III
Period Duration	168 hours (7 days)	168 hours (7 days)	4 hours	504 hours (21 days)	4 hours	72 hours (3 days)	72 hours (3 days)
Temperature (ambient and test fluid)	158° ±2°F	Room	158° ± 2°F	Room	158° ± 2°F	—67° ± 2°F	130° ± 5°F
Operation and tests during period	None	Actuate valve at least twice per day	None	Actuate valve at least twice per day	None	None	None
Operation and tests immediately after period	(1) Connect nozzle to adapter; actuate 5 times and diconnect nozzle. (2) Conduct Leakage test using MIL-S-3136, type I, fluid. There shall be no leakage after 15 minutes.	Conduct Leakage test using MIL-S-3136, type III, fluid.	(1) Connect nozzle to adapter; actuate 5 times, and disconnect nozzle. (2) Conduct Leakage test using MIL-S-3136, type I, fluid. There shall be no leakage after 15 minutes.	Conduct Leakage test using MIL-S-3136, type I, fluid.	(1) Connect nozzle to adapter; actuate 5 times and disconnect nozzle. (2) Conduct Leakage test using MIL-S-3136, type I, fluid. There shall be no leakage after 15 minutes.	With ambient temperature and test fluid at not higher than —65°F: (1) Connect nozzle to adapter, actuate 5 times, and disconnect nozzle. (2) Conduct Leakage test using MIL-S-3136, type I, fluid.	With ambient temperature and test fluid at 130° ±5°F: (1) Connect nozzle to adapter, actuate 5 times, and disconnect nozzle. (2) Conduct Leakage test using MIL-S-3136, type III, fluid.

Notes: (a) During periods of soaking in the test fluid, the adapter shall be mantained is such a manner as to insure complete contact of all nonmetallic parts with the fluid as would be expected under service conditions.

(b) Each period shall follow immediately after the preceding one in the order noted.

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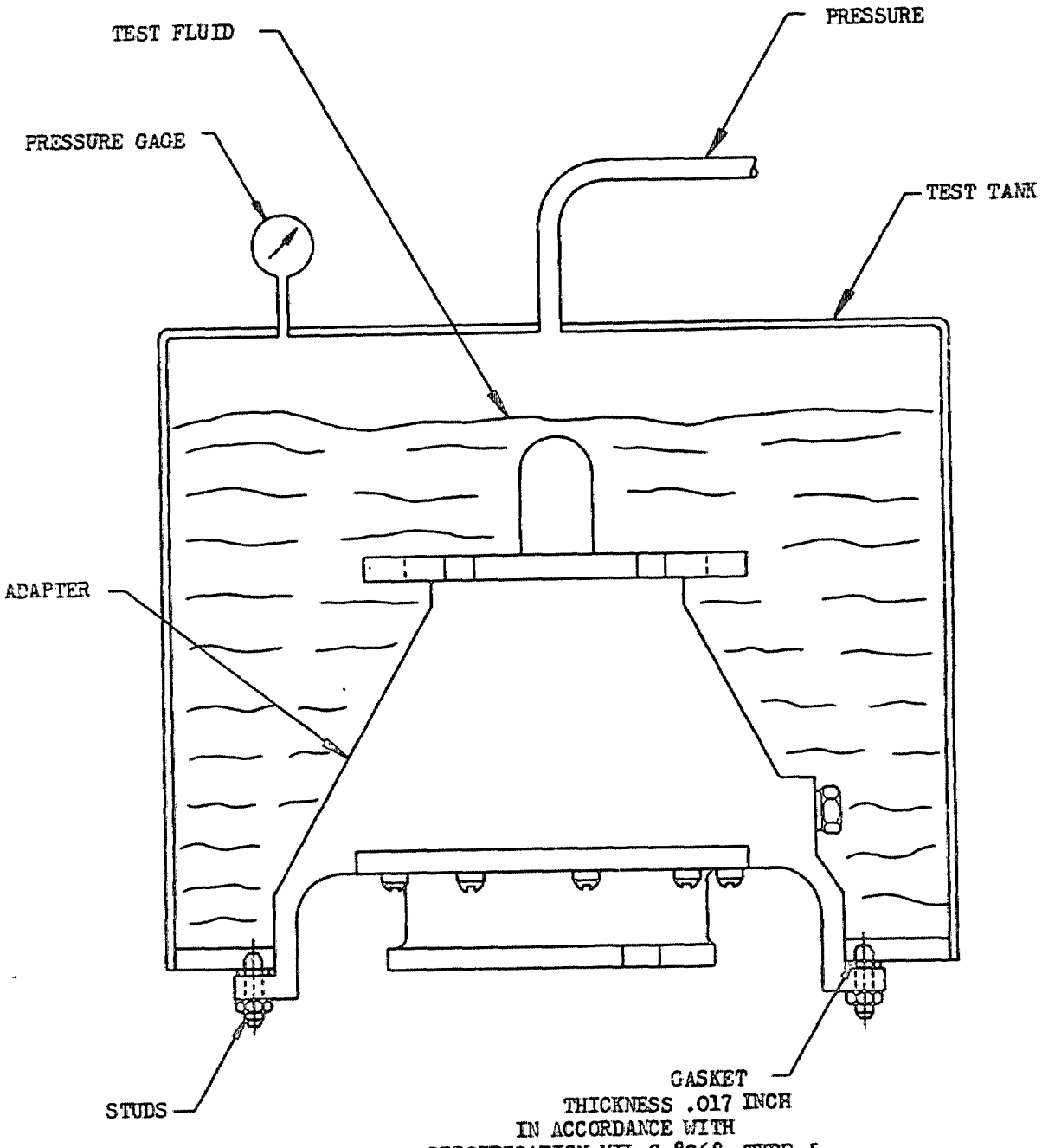


FIGURE 2. *Tank Leakage test setup for type A-2 and type A-6 adapters*

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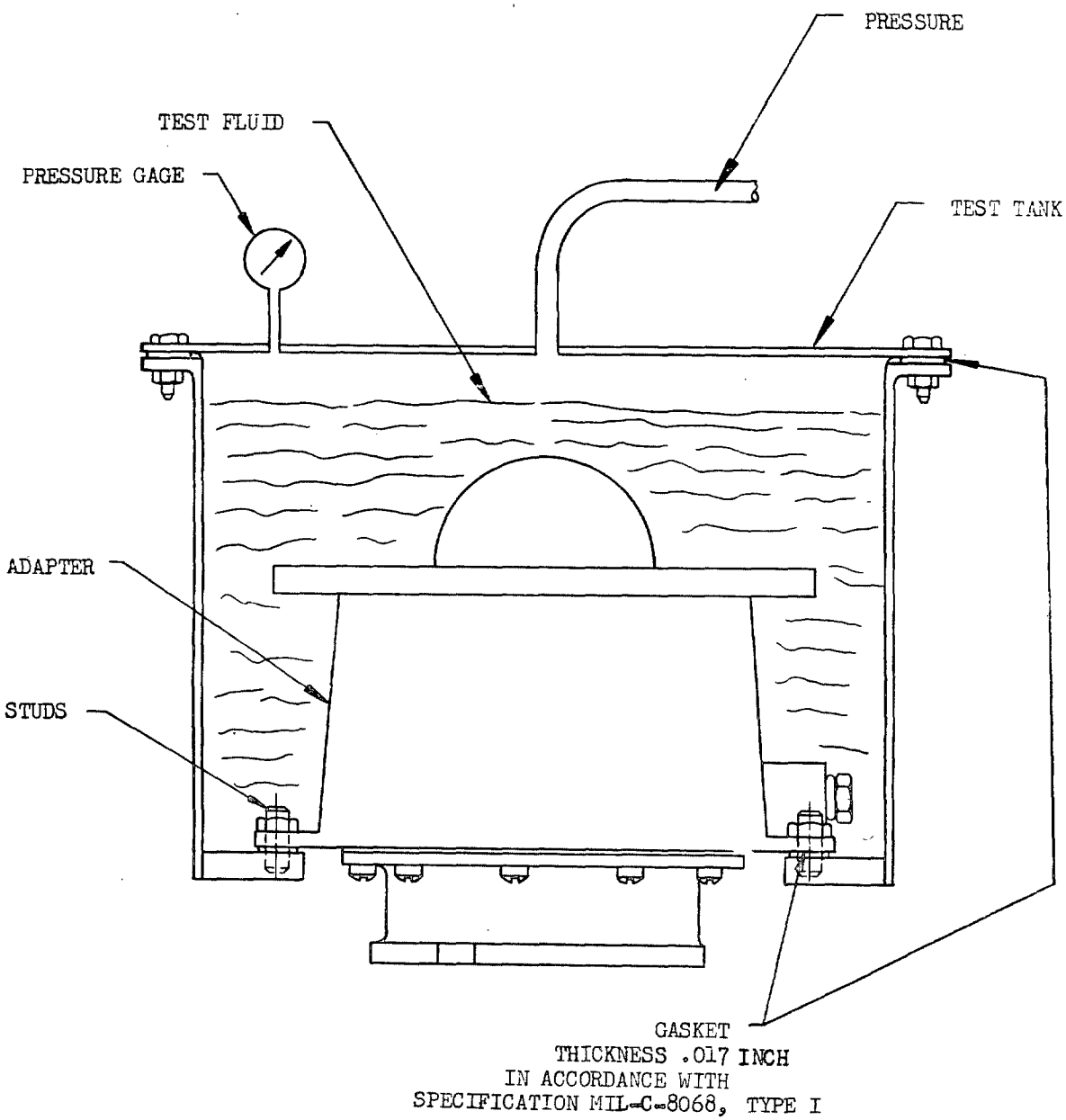


FIGURE 3. *Tank Leakage test setup for type A-3 and type A-4 adapters*

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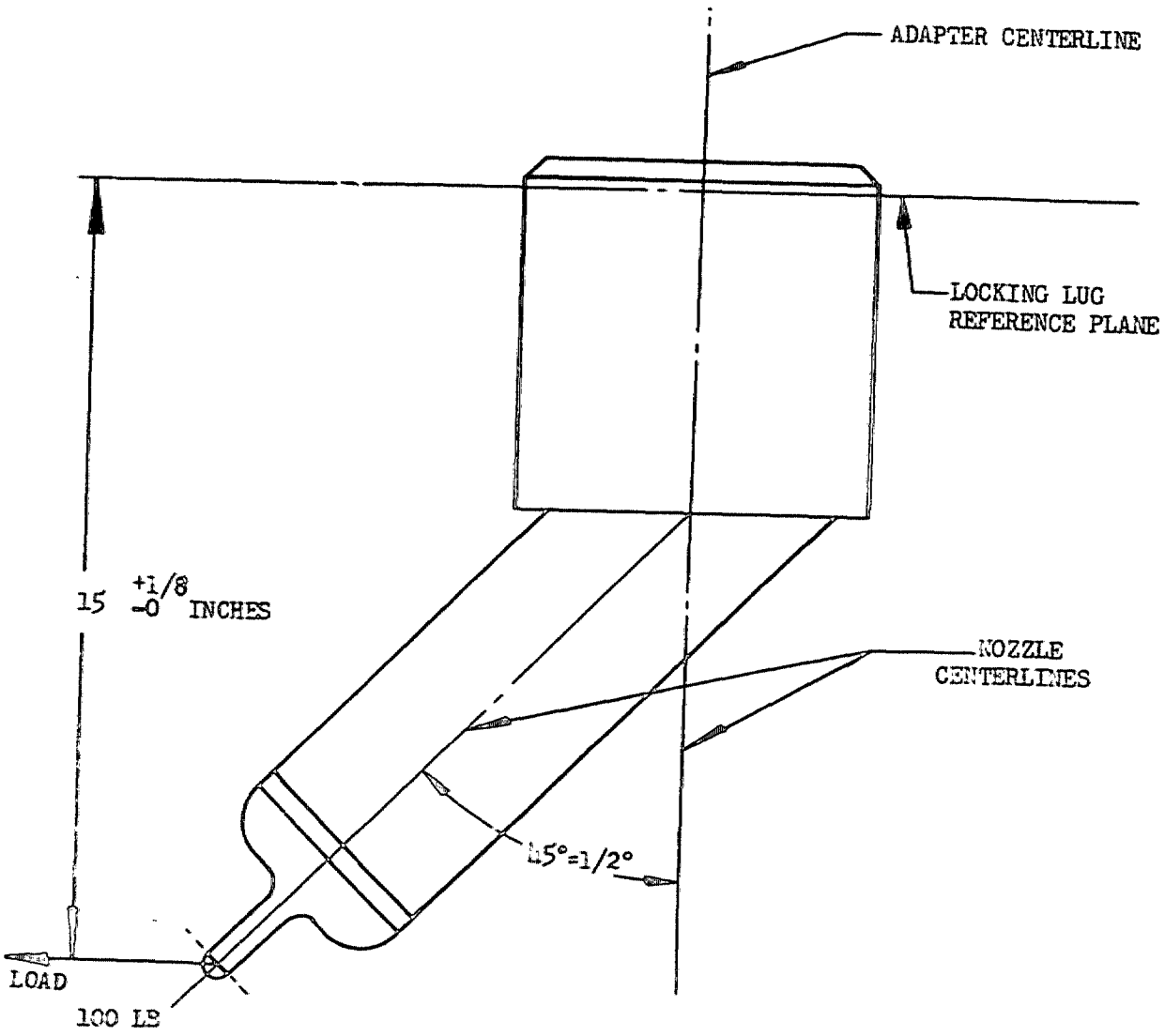


FIGURE 4. *Load application for Strength test*

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4.5.8 *Vibration.* The adapter shall be adequately mounted on the vibration device and subjected to the four vibration-scanning cycle tests listed in table II. The test shall be conducted at room temperature. There shall be no evidence of leakage during the test, damage or loosening of parts, or leakage as a result of the test.

TABLE II. *Vibration test*

Scanning cycle test	1	2	3	4
Axis of vibration	Along axis of symmetry	Along axis of symmetry	Normal to axis of symmetry	Normal to axis of symmetry
Fluid pressure	60 psi	½ psi	60 psi	½ psi
Scanning cycle time	15 min	15 min	15 min	15 min
No. of scanning cycles per test	2	2	2	2
Procedure	The frequency shall be uniformly increased with respect to time through a frequency range from 10 to 500 cps with an applied double amplitude of 0.036 inch up to 75 cps, and from there an applied vibration acceleration not less than ±10g. The frequency shall be similarly decreased in such a manner that the complete cycle is accomplished in the specified cycle time.			

4.5.9 *Proof pressure.* The adapter shall be subjected to a fluid-proof pressure of 120 ±2 psi for a minimum period of one minute. The pressure shall be applied to the adapter outlet. There shall be no evidence of leakage, distortion, or other injury to any part of the assembly.

4.5.10 *Burst pressure.* The adapter shall be subjected to a fluid pressure of 180 ±2 psi for a minimum period of 1 minute. The pressure shall be applied to the adapter outlet. There shall be no evidence of distortion or other injury to any part of the adapter, and, when the pressure is lowered to 60 psi, there shall be no evidence of leakage from any part of the adapter.

4.5.11 *Contaminated-fuel endurance.* Fuel, containing a quantity and type of contaminant conforming to table III for every 20 gallons, shall be pumped through the nozzle and adapter in the connected position in a recirculating system which circulates fuel at a rate of 200 gpm. The units shall be operated 200 cycles. A cycle shall consist of operating the flow-control handle of the nozzle from the "closed" position to the "open" position, allowing the units to remain in this position for a minimum period of 1 minute and returning the flow-control handle to the "closed" position. The test fuel shall be properly agitated to keep the contaminant uniformly distributed in the circulating fuel. After this test, the unit shall be flushed out with a clear fuel and drained, following which, the adapter shall be subjected to the Leakage test (4.5.2.3).

4.5.12 *Disassembly and inspection.* The unit shall be disassembled and inspected. If corrosion, deterioration, or wear exists to a degree which affects performance, the unit shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 **Application.** The requirements of section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 **Levels of preservation and packaging.**

5.2.1 *Level A.* Adapters shall be wrapped in kraft paper conforming to Specification UU–P–271 and placed in containers conforming to Specification MIL–B–4229, PPP–B–566, PPP–B–676, or PPP–B–636. Unit quan-

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TABLE III. *Contaminated fuel endurance test dust*

Contamination description	Particle size	Quantity (grams)
Sharp silica sand	Passes through 30 mesh screen held on 50 mesh screen.	0.20
Sharp silica sand	Passes through 50 mesh screen held on 100 mesh screen.	0.20
Prepared Dust AC Spark Plug Co. Package No. 1543637, or equal.	0–5 microns 12 ±2 percent 5–10 microns 12 ±3 percent 10–20 microns 14 ±3 percent 20–40 microns 23 ±3 percent 40–80 microns 30 ±3 percent 80–200 microns 9 ±3 percent	1.6

tities shall be as specified by the procuring activity.

5.2.2 *Level C.* Adapters shall be preserved and packaged in accordance with manufacturer's commercial practice.

5.3 Levels of packing.

5.3.1 *Level A.* Adapters, preserved and packaged in accordance with 5.2.1, shall be packed in exterior type shipping containers conforming to Specification MIL–B–138, PPP–B–585, PPP – B – 591, PPP – B – 621, PPP–B–601, PPP–B–636, type I, class 3, or type II, class 3, or Specification MIL–B–10377. As far as practicable, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to Specification MIL–L–10547 and shall be sealed in accordance with the appendix thereto. The case liner will not be required when the unit, intermediate, or exterior container conforms to Specification PPP–B–636, type I, class 3 or type II, class 3, and is sealed at all joints and seams, including manufacturer's joint, with tape conforming to Specification PPP–T–60.

5.3.2 *Level B.* Adapters, preserved and packaged in accordance with 5.2.1, shall be packed in domestic-type exterior containers conforming to Specification PPP–B–591, PPP–B–601, PPP–B–585, PPP – B – 621, PPP–B–636, or MIL–B–10377. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shape and size and contain identical quantities. The gross weight of each pack shall be limited to 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When fiberboard containers are used, the fiberboard shall meet the special requirements table of Specification PPP–B–636.

5.3.3 *Level C.* Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or regulations of other common carriers as applicable to the mode of Transportation.

5.4 Marking. Interior and exterior containers shall be marked in accordance with Standard MIL–STD–129.

5.4.1 The following note shall be applied on unit and intermediate packages and shipping containers:

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IF IN STORAGE AFTER (DATE)*, THE ADAPTER SHALL BE TESTED AND INSPECTED BEFORE USE.

*Note: Insert date 6 quarters after assembly date of the adapter.

6. NOTES

6.1 Intended use. The adapters covered by this specification are intended for use as the airplane fuel-system filling connection to be used in conjunction with pressure fuel servicing nozzles on airplanes using only hydrocarbon fuel.

6.2 Definitions

6.2.1 *Cycle.* Unless otherwise specified, a cycle shall be defined as the connecting of the test nozzle to the adapter, fully opening and closing the flow control valve four times, and removing the test nozzle from the adapter.

6.2.2 *Inlet.* For purposes of this specification the bayonet flange end of the adapter is called the adapter inlet.

6.3 Ordering data. Requisitions, contracts, and orders should specify whether overseas packing is required.

6.4 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Bureau of Aeronautics, Navy Department, Washington 25, D.C.

6.5 International standardization. Provisions of this specification are the subject of international standardization agreements. When amendments, revisions, or cancellation of this specification is proposed, the departmental custodian will inform their respective Departmental Standardization Office (DepSO) so that appropriate action may be taken respecting the international agreement concerned.

Notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Ordnance Corps
Navy—Bureau of Aeronautics
Air Force

Other interest:
International (See 6.5)

Preparing activity:
Navy—Bureau of Aeronautics

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD, FORM 1 APR 65 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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